CHEATSHEET · QUICK REFERENCE

Lab 2 — Storage Deployment: RAID 0/1/5/6/10 with mdadm

Build every RAID level the exam tests on loopback disks with mdadm, then fail and rebuild an array to learn capacity and fault-tolerance formulas.

Create arrays with mdadm

```
mdadm --create /dev/md0 --level=0 --raid-devices=2 /dev/loop1 /dev/loop2
```
RAID 0 stripe

```
mdadm --create /dev/md1 --level=1 --raid-devices=2 /dev/loop1 /dev/loop2
```
RAID 1 mirror

```
mdadm --create /dev/md5 --level=5 --raid-devices=3 /dev/loop1 /dev/loop2 /dev/loop3
```
RAID 5 single parity

```
mdadm --create /dev/md6 --level=6 --raid-devices=4 /dev/loop1 /dev/loop2 /dev/loop3 /dev/loop4
```
RAID 6 double parity

```
mdadm --create /dev/md10 --level=10 --raid-devices=4 /dev/loop1 /dev/loop2 /dev/loop3 /dev/loop4
```
RAID 10 stripe of mirrors

```
mdadm --detail /dev/md5 | grep -E "Array Size|Used Dev Size"
```
Confirm usable capacity

Capacity and fault-tolerance formulas

RAID	Usable capacity	Fault tolerance
0	N x disk	0 (none)
1	1 x disk (50% overhead)	1 disk
5	(N-1) x disk	1 disk
6	(N-2) x disk	2 disks
10	N / 2 x disk	1 disk per mirror
JBOD (linear)	sum of disks	0 (none)

Fail and rebuild

```
mdadm /dev/md1 --fail /dev/loop2 --remove /dev/loop2
```
Mark a disk failed and remove it

```
mdadm /dev/md1 --add /dev/loop3
```
Add a replacement / spare

```
watch -n1 cat /proc/mdstat
```
Observe the resync percentage

Hardware vs. software RAID

Factor	Hardware RAID	Software (mdadm)
CPU cost	None (on controller)	Host CPU
Battery-backed cache	Yes (BBWC/FBWC)	No
Portability	Tied to controller	Any Linux host
Cost	Hundreds of $	Free